

$\text{var } x, n: \text{Int}$
 $\text{con } N: \text{Int}$
 $\{N \geq 0\}$
 $x, n := 1, 0;$
 $\{i\}$
 $\text{do } n \neq N \rightarrow$

σ_x	σ_n
$0! = 1$	0
$1! = 1$	1
$2! = 2$	2
$3! = 6$	3
$\vdots$	$\vdots$
	N

$x, n := x \times (n+1), n+1$

$\{1 \wedge n \neq N\}$
 $\{x = N!\}$

1. $x = n! \wedge 0 \leq n \leq N$

2. -

Hypo $P = I$

$\{N \geq 0\} \equiv N \geq 0 \Rightarrow \text{wp.}(x, n := 1, 0). I$

$x, n := 1, 0;$

$\{i\}$

Sup $N \geq 0$

$\text{wp.}(x, n := 1, 0). (x = n! \wedge 0 \leq n \leq N)$

$\equiv \{Def + \text{wp}\}$

$1 = 0! \wedge 0 \leq 0 \leq N$

$\equiv \{Arithmetic\}$

$1 = 1$ $\wedge$ $0 \leq N$

$\equiv \{Arith., Sup\}$

True

3.-

$$\underbrace{I \wedge n \neq N} \Rightarrow x = N!$$

$$\sup x = n! \wedge 0 \leq n \leq N \wedge n = N$$

$$\begin{aligned} & x = N! \\ & \equiv \{ \sup n = N \} \\ & x = n! \\ & \{ \sup x = n! \} \end{aligned}$$

True

4.-

$$\{ I \wedge n \neq N \}$$

$$x, n := x * (n+1), n+1 \equiv$$

$$\{ I \}$$

$$I \wedge n \neq N$$

$$\Rightarrow \text{wp} \cdot (x, n := x * (n+1), n+1) \cdot I$$

$$\sup x = n! \wedge 0 \leq n \leq N \wedge n \neq N$$

$$\text{wp} \cdot (x, n := x * (n+1), n+1) \cdot (x = n! \wedge 0 \leq n \leq N)$$

$$\equiv \{ \text{Def wp} \}$$

$$x * (n+1) = (n+1)! \wedge \underline{0 \leq n+1 \leq N}$$

$$\equiv \{ \text{Arithm} \}$$

$$x * (n+1) = \underline{(n+1)!}$$

$$\equiv \{ \text{Def !} \}$$

$$x * (n+1) = (n+1) * n!$$

$$\equiv \{ \sup n \geq 0, \text{Arithm} \}$$

$$x = n!$$

$$\equiv \{ \sup x = n! \}$$

True

$$\{ : N - n$$

$$(i) \quad I, n \neq N \Rightarrow N - n \geq 0$$

$$\sup x \geq n! \wedge 0 \leq n \leq N \wedge n \neq N$$

$$N - n \geq 0$$

$\equiv$
 $\vdots$
 $\equiv$

True

Ejercicio 1

(ii)

$$\{ I, n \neq N, N - n = T \}$$

$$x, n := x * (n+1), n+1$$

$$\{ N - n < T \}$$

$$\equiv I, n \neq N, N - n = T$$

$$\Rightarrow \text{wp. } (x, n := x * (n+1), n+1) . (N - n \leq T)$$

$$\sup x \geq n! \wedge 0 \leq n \leq N, n \neq N, N - n = T$$

$$\text{wp. } (x, n := x * (n+1), n+1) . (N - n \leq T)$$

$$\equiv \{ \text{Def wp} \}$$

$$N - (n+1) \leq T$$

$$\equiv \{ \text{Arithm} \}$$

$$N - n - 1 \leq T$$

$$\equiv \{ \sup N - n = T \}$$

$$T - 1 \leq T$$

$$\equiv \{ \text{Arithm} \}$$

True